Coripet RVM Prospective Predictor

How AI helps decide where to place bottle-recycling machines, explained for everyone

Production Decision Support Dashboard | System Manual Overview

INTRODUCTION

Where should the next machine go?

An AI tool that estimates how many bottles a new machine will collect, before it is installed.



Predict

Forecast visitors and bottles for any new store



Rank

Compare candidate stores and pick the best first



Monitor

Spot problems in the 1,849 running machines



In simple words: the right machine, in the right store, before any money is spent.

KEY IDEA

What is machine learning?

Definition: teaching a computer to make predictions by showing it many real examples, instead of writing the rules by hand.



1. Learn from the past

1.48 million real days from 1,681 stores



2. Find patterns

e.g. bigger stores and busy towns → more bottles



3. Predict

Estimate bottles for a new store



In simple words: like an experienced manager who has seen thousands of stores and can guess well how a new one will do.

Evaluate a store in four simple steps

STEP 1

Location

Store name and address

STEP 2

Store size

Floor area and checkouts

STEP 3

Reward & date

Reward and start month

STEP 4

Forecast

Full results for the store

HOW THE LOCATION IS FOUND

1. Exact postal code + town match
2. Town centre on the map
3. Town registry lookup
4. National average (last resort)

ADDED AUTOMATICALLY

- Population
- Area (km²)
- People per km²
- Town GDP (PIL)
- GDP per person
- Store size (25,104 stores)

In simple words: type the address; the app finds the town, adds its facts and the store size by itself.

TAB 1 · STEP 3

What do shoppers get for their bottles?

Loyalty points

+35–45%

more visitors

Points on the store card. Strongest habit.

Discount voucher

+15–25%

more bottles

e.g. €3 off a €25 shop per 200 bottles.

Free gift

+10–18%

more bottles

A direct prize, no minimum spend.

Custom policy

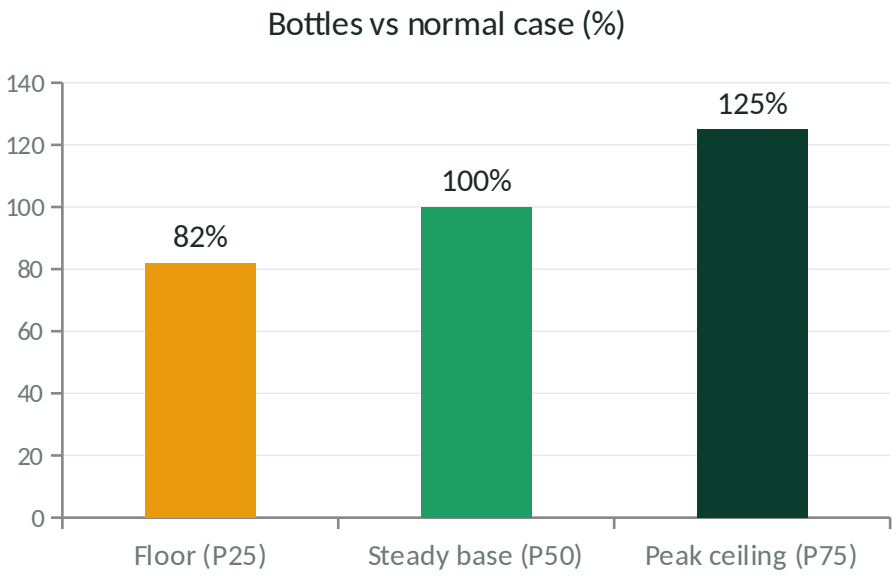
You decide

set your own rules

Discount, minimum spend, bottles needed.

In simple words: a better reward brings more people back with more bottles; loyalty points work best.

Three scenarios, not just one guess



Suitability gauge

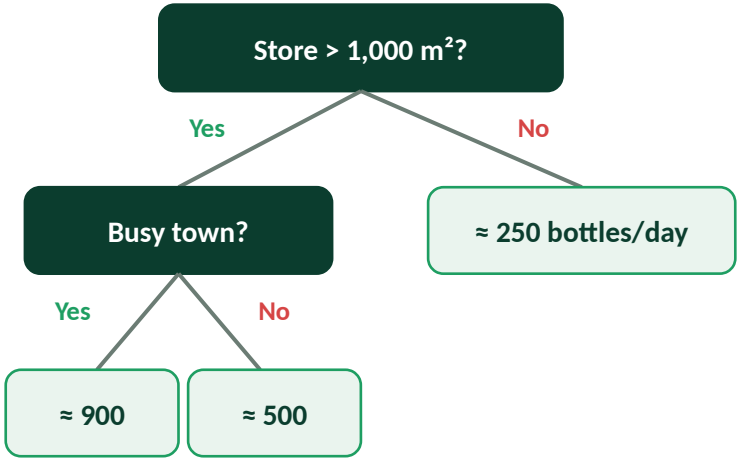
A 0–100% dial showing how good the site is compared with the whole network.

Location analysis text

Every key number in one copy-ready block for reports and memos.

In simple words: we plan for a bad month, a normal month and a great month, not a single magic number.

What is LightGBM?



One simple tree (example numbers). LightGBM adds hundreds of these together.

Decision tree
A flowchart of yes/no questions that ends in a number.

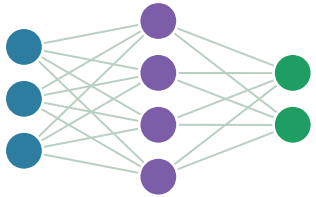
Boosting
Trees are built one after another; each new one fixes the mistakes the earlier ones still make.

Gradient
The maths that tells each new tree which way to correct, and by how much.

Light
Fast and light on memory (made by Microsoft). Excellent for table data like ours.

In simple words: a team of simple flowcharts, each fixing what the others got wrong; it gives 90% of the bottle forecast.

What is a neural network?



A neural network is a chain of layers of tiny calculators (“neurons”), loosely inspired by brain cells passing signals. Ours is built with PyTorch.

Neuron
Multiplies each input by a weight, adds them up, passes it on.

Layer
A row of neurons. Ours: 256 → 128 → 64, from simple facts to bigger ideas.

Weights
Adjustable knobs. Training = turning them so mistakes get smaller.

Activation (GELU)
A smooth switch: lets the network learn curves, not just straight lines.

Dropout
Randomly switches off 10–20% of neurons while training, so it can't memorise.

LayerNorm
Keeps numbers in each layer on a similar scale, so learning stays stable.

In simple words: many tiny calculators in layers; training turns their knobs until the predictions get close to reality.

How the AI learns: practice, check, improve

1. Guess

Predict bottles for a real machine on a real past day

2. Compare

Put the guess next to what really happened

3. Measure

Score how wrong it was (the “loss”)

4. Adjust

Tweak its internal numbers a little to do better

Repeat many times
over 1.48 million real daily records

Final exam
tested on data it never saw, so it can't just memorise

In simple words: like a student doing practice tests: guess, check the answer, learn from the mistake, repeat.

Two AI “brains” working as a team

LightGBM (decision trees)

Hundreds of small trees ask yes/no questions, like “20 questions”.

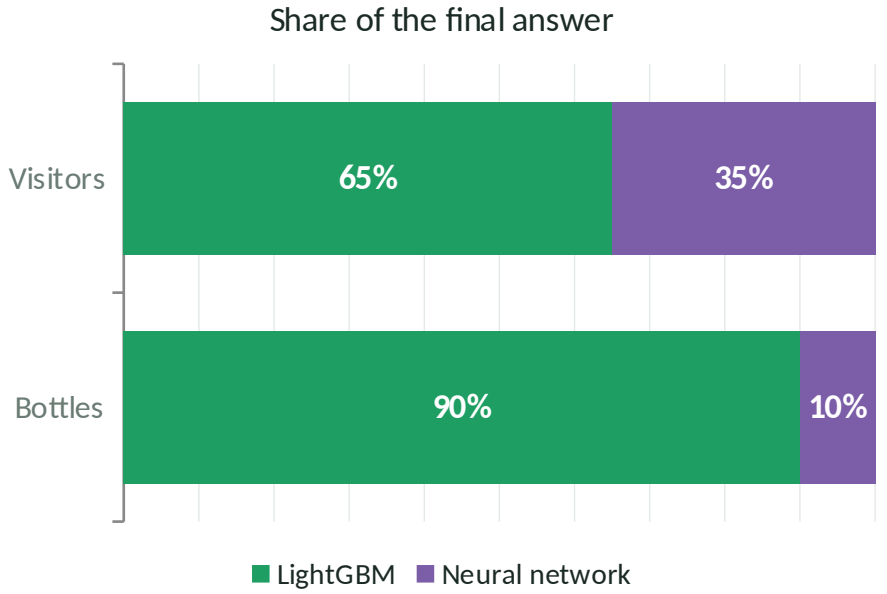
e.g. Store > 1,000 m²? Yes → Dense town? Yes → about 800 bottles/day

Neural network

Loosely inspired by the brain. Spots subtle mixes of factors.

e.g. this brand + big city + summer = extra visitors

Why mix them? They make different mistakes, so blending cancels some out.



In simple words: two experts give their opinion, and we combine them with the weights that proved most accurate.

Learning without being fooled by strange days

The problem: strange days

A restaurant drops off 8,000 bottles in one day

A broken machine collects 0 bottles for days

If the AI copied these days, its forecasts for normal days would be wrong.

How the AI stays calm

Strange days get less attention
One unusual day can't undo what thousands of normal days taught it.

Small and big stores count equally
Very busy stores can't drown out the smaller ones.

Broken machines are trusted less
Data from faulty machines counts much less.

In simple words: like a good teacher who does not judge a student on one strange exam day.

Example: one strange day

Example: a store normally collects about 800 bottles a day. One day a restaurant drops off 8,000.

1

Keep it in proportion

The jump is compared with the store's normal level, so it looks less extreme.

2

Limit the push

No single day can change the AI more than a set limit.

3

Trust broken machines less

If the machine is known to be faulty, the day counts only ¼.

How much one day can change the AI

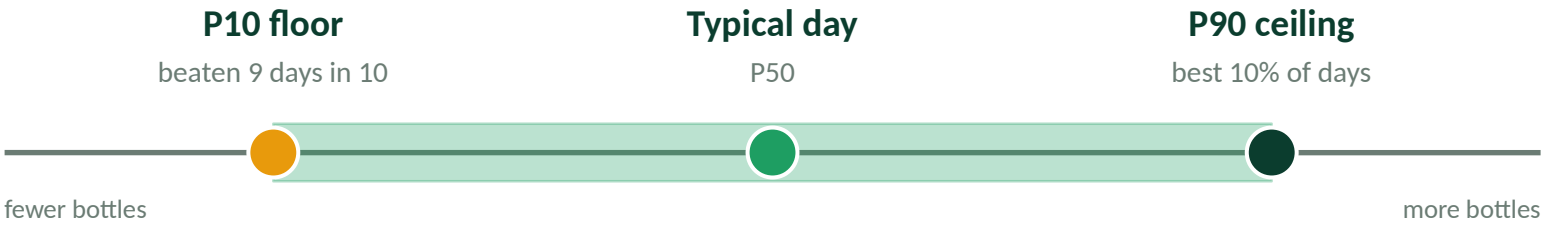


Result: the AI notices the day, but its forecast only moves a little.

In simple words: one strange day is noticed, but it cannot change the forecast much.

RESULTS · RISK

How risky is this investment?



Capital safety

Chance of beating 250 bottles/day breakeven

Capacity check

Overflow risk > 20% → warning to upgrade machine

Past vs forecast

Installed machines: 24 months of real data vs 12 months forecast

In simple words: we show the bad case, the typical case and the great case, plus the odds of making money.

Six safety gates before saying “yes”

1. No machine too close

Warn < 750 m, fail < 250 m

2. Breaks even

At least 250 bottles per day

3. Store big enough

≥ 300 m² and ≥ 3 checkouts

4. Enough people

≥ 5,000 residents, ≥ 150 per km²

5. Right-sized machine

Not too small, not oversized

6. Easy to service

Within ~25 km of a depot

All pass
Fully approved

Some warnings
Conditional: optimise first

A gate fails
Conditional / rejected

In simple words: a checklist like an airport security gate; green means go, amber means fix first, red means stop.

TABS 2 & 3

SPL and POT: live lists of candidate stores

SPL Machines

= Sopralluogo Completato

61 stores, ready for deployment planning.

POT Machines

= Potenziale

331 sites, screened on paper before a visit.

Live data from Coripet's system, with a backup copy

Only real shops kept (no test sites)

Survey recency filter

- **Last 5 months (top priority)**
- Last 12 months
- Last 24 months
- All sites

In simple words: two to-do lists of possible new stores, always up to date, with the freshest ones first.

TAB 4

Spotting machines in trouble

Offline
Zero bottles 7+ days

→ Send a technician

Bin full
Visitors normal, bottles low

→ Empty bags twice a day

Up & down
Big jump, then a crash

→ Check expired promotion

Huge spike
> 7,500 bottles in a day

→ Check bulk drops

Underperformer
< 250/day vs high forecast

→ Move to the entrance

Super hub
> 3,000 bottles/day

→ Add a 2nd machine

Sensor drift
1–5 bottles, 0 users, 14 days

→ Clean & recalibrate

Healthy
No rules triggered

→ Normal service

In simple words: a doctor for the fleet: it spots the symptom and suggests the treatment.



Thank you

Questions and discussion are welcome